Second-Order ODEs

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Homogeneous Equations with Constant Coefficients

• We emphasize once again that there are no general methods for solving second (or higher) order linear differential equations. However, there are some special cases for which solution methods do exist. We will consider such a case, linear equations with constant coefficients. First, we treat homogeneous equations; nonhomogeneous equations will be treated later.
• A second-order linear homogeneous differential equation with constant coefficients is an equation which can be written in the form

linear homogenen 2ndorde ODE with constant coeff —

$$y'' + ay' + by = 0 \tag{1}$$

where a and b are real numbers.
• We have already seen that the function $y = \mathrm{e}^{-ax}$ is a solution of the first-order linear equation

$$y' + ay = 0 \text{ (the model for exponential growth and decay).}$$

Homogeneous Equations with Constant Coefficients. Contd.

- The previous observation hints us about the possibility that (1) may also have an exponential function $y = \mathrm{e}^{rx}$ as a solution.
- If $y = \mathrm{e}^{rx}$, then $y' = r\mathrm{e}^{rx}$ and $y'' = r^2\mathrm{e}^{rx}$. Now substitution of these identities into (1) gives

$$r^2\mathrm{e}^{rx} + a(r\mathrm{e}^{rx}) + b(\mathrm{e}^{rx}) = \mathrm{e}^{rx}(r^2 + ar + b) = 0.$$

- Since $\mathrm{e}^{rx} \neq 0$ for all x, we conclude that $y = \mathrm{e}^{rx}$ is a solution of (1) if and only if

$$r^2 + ar + b = 0. \tag{2}$$

- Thus, if r is a root of the quadratic equation (2), then $y = \mathrm{e}^{rx}$ is a solution of equation (1); we can find solutions of (1) by finding the roots of the quadratic equation (2).

Homogeneous Equations with Constant Coefficients. Contd.

Definition (Characteristic Polynomial and Roots)

Given the differential equation (1). The corresponding quadratic equation

$$r^2 + ar + b = 0$$

is called the *characteristic equation* of (1); the quadratic polynomial $r^2 + ar + b$ is called the *characteristic polynomial*. The roots of the characteristic equation are called the *characteristic roots*.

r_1, r_2

Nature of Solutions

• The nature of the solutions of the differential equation (1) depends on the nature of the roots of its characteristic equation (2). There are three cases to consider:

(1) Equation (2) has two, distinct real roots, $r_1 = \alpha, r_2 = \beta$.

(2) Equation (2) has only one real root (repeated), $r = \alpha, \alpha$

(3) Equation (2) has complex conjugate roots, $r_1 = \alpha + i\beta$, $r_2 = \alpha - i\beta$, $\beta \neq 0$.

Case	Roots	General Solution
Distinct Real	$r_1 \neq r_2$	$y = C_1 e^{r_1 x} + C_2 e^{r_2 x}$
Repeated Real	$r_1 = r_2 = r$	$y = (C_1 + C_2 x) e^{rx}$
Complex Roots	$r = \alpha \pm i\beta$	$y = e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x)$

check the Wronskian.

$y_1(x) = e^{r_1 x}$

$y_2(x) = e^{r_2 x}$

$$W[y_1, y_2](x) = \begin{vmatrix} e^{r_1 x} & e^{r_2 x} \\ r_1 e^{r_1 x} & r_2 e^{r_2 x} \end{vmatrix}$$

$= (r_2 - r_1) e^{r_1 x} e^{r_2 x}$

$(r_1 \neq r_2)$ $\forall x$ $0 \neq$

Discriminant Method

Let $D = b^2 - 4ac$

- If $D > 0$: Real and distinct roots

$$y = C_1 e^{r_1 x} + C_2 e^{r_2 x}$$

- If $D = 0$: Repeated root

$$y = (C_1 + C_2 x)e^{rx}$$

- If $D < 0$: Complex roots

$$y = e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x)$$

Example

Example (Linearly Independent Solutions)

Find a linearly independent pair of solutions of

$$y'' + 3y = 0,$$

and give the general solution of the equation.

$$\boxed{y = e^{rx}}$$

Characteristic eqn : $r^2 + 3 = 0$.

$r = 0 \pm \sqrt{3}\, i \Rightarrow \alpha = 0, \; \beta = \sqrt{3}$.

$y(x) = c_1 \cos(\sqrt{3}x) + c_2 \sin(\sqrt{3}x)$

$W[y_1, y_2](x) = \sqrt{3}$.

$y_1(x) = \cos(\sqrt{3}x)$

$y_2(x) = \sin(\sqrt{3}x)$

Example (Linearly Independent Solutions)

Find the general solution of the differential equation

$$y'' - 6y' + 9y = 0.$$

$y = e^{rx}$.

EXR

The characteristic equation is : $r^2 - 6r + 9 = 0$

The roots are $r = 3, 3$.

The general sol^n is $y(x) = (c_1 + c_2 x)e^{3x}$.

IVP for Second-Order Homogeneous Linear ODEs

Example

Find the solution of the initial-value problem:

$$y'' + 2y' - 15y = 0, \; y(0) = 2, \; y'(0) = -6.$$

general solⁿ $y(x) = c_1 y_1(x) + c_2 y_2(x)$

use the auxiliary condⁿs.

(EXR)

Euler Equation

Definition (Euler Equations)

A second-order linear homogeneous equation of the form

$$x^2\frac{d^2y}{dx^2} + \alpha x\frac{dy}{dx} + \beta y = 0 \tag{3}$$

where α and β are constants, is called an Euler equation.

Euler equation (3) can be transformed into the second order equation with constant coefficients

$$\frac{d^2y}{dz^2} + a\frac{dy}{dz} + by = 0 \longrightarrow \text{constant coefficients}$$

where a and b are constants, by means of the change of independent variable $\boxed{z = \ln x}$.

Since $z = \ln x$. By diffentiating y in terms of 'x', we get

$$\frac{dy}{dx} = \frac{dy}{dz}\frac{dz}{dx} = \frac{1}{x}\frac{dy}{dz}.$$

Again differentiating, we get

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{1}{x}\frac{dy}{dz}\right) = \frac{1}{x}\frac{d}{dx}\left(\frac{dy}{dz}\right) + \frac{d}{dx}\left(\frac{1}{x}\right)\frac{dy}{dz}.$$

$$= \frac{1}{x}\frac{d}{dz}\left(\frac{dy}{dz}\right)\frac{dz}{dx} - \frac{1}{x^2}\frac{dy}{dz}.$$

$$= \frac{1}{x}\frac{d^2y}{dz^2}\frac{dz}{dx} - \frac{1}{x^2}\frac{dy}{dz} = \frac{1}{x^2}\frac{d^2y}{dz^2} - \frac{1}{x^2}\frac{dy}{dz}$$

$$\Rightarrow x\frac{dy}{dx} = \frac{dy}{dz} \quad \text{and} \quad x^2\frac{d^2y}{dx^2} = \frac{d^2y}{dz^2} - \frac{dy}{dz}.$$

Substituting these in the main equation, we get

$$\frac{d^2y}{dz^2} - \frac{dy}{dz} + \alpha\left(\frac{dy}{dz}\right) + \beta y = 0$$

$$\Rightarrow \frac{d^2y}{dz^2} + (\alpha-1)\frac{dy}{dz} + \beta y = 0$$

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or $$\frac{d^2y}{dz^2} + \alpha'\frac{dy}{dz} + \beta' y = 0$$

where $\alpha' = \alpha - 1$ and $\beta' = \beta$.

Solution of Euler Equation

Example

Find the general solution of the Euler equations

$$x^2y'' - xy' + 5y = 0.$$

Under the transformation $z = \ln x$ the above equation reduces to

$$\frac{d^2y}{dz^2} + (-1-1)\frac{dy}{dz} + 5y = 0$$

$$\Rightarrow \frac{d^2y}{dz^2} - 2\frac{dy}{dz} + 5y = 0$$

— characteristic polynomial is $r^2 - 2r + 5 = 0$

Characteristics roots are

$$r = \frac{2 \pm \sqrt{4 - 4.5}}{2}$$

$$= \frac{2 \pm 4i}{2} = 1 \pm 2i$$

Hence the general sol^n is $y(z) = e^z(c_1 \cos 2z + c_2 \sin 2z)$

Now going back to the 'x' variable:

$$y(x) = e^{\ln x}\left(c_1 \cos(2 \ln x) + c_2 \sin(2 \ln x)\right)$$

- Also, the Euler equ^n (sometimes, it is also referred as Cauchy-Euler equ^n)

$$\underbrace{ax^2y'' + bxy' + cy = 0}$$

can be solved by making the guess: $\boxed{y = x^r}$

$\Rightarrow \underbrace{(ar(r-1)+br+c)}x^r = 0$. by solving the characteristic equ^n $ar(r-1)+br+c=0$, we get

Case	Roots	General Sol^n
Distinct roots	$r_1 \neq r_2$	$y(x) = c_1 x^{r_1} + c_2 x^{r_2}$
Real repeated roots	$r_1 = r_2 = r$	$y(x) = (c_1 + c_2 \ln\lvert x\rvert)x^r$
Complex (conjugates)	$r_1 = \alpha + i\beta$ $r_2 = \alpha - i\beta$	$y(x) = x^\alpha(c_1\cos(\beta\ln\lvert x\rvert) + c_2\sin(\beta\ln\lvert x\rvert))$

Nonhomogeneous Equation

• Now we will consider the general second-order linear nonhomogeneous equation

$$y'' + p(x)y' + q(x)y = f(x) \tag{4}$$

where p, q, f are continuous functions on an interval I.

• The objectives of this section are to determine the "structure" of the set of solutions of (4) and to develop a method for constructing a solution of (N) using two linearly independent solutions of the corresponding homogeneous equation

$$y'' + p(x)y' + q(x)y = 0. \tag{5}$$

• As we shall see, there is a close connection between equations (4) and (5). In this context, equation (5) is called the reduced equation of equation (4).

Theorem

If $z = z_1(x)$ and $z = z_2(x)$ are solutions of equation (4), then

$$y(x) = z_1(x) - z_2(x)$$

is a solution of equation (5).

Remark

The above theorem says that the difference of any two solutions of the nonhomogeneous equation (4) is a solution of its reduced equation (5).

$$z_1'' + p(x) z_1' + q(x) z_1 = f(x) = z_2'' + p(x) z_2' + q(x) z_2$$

$$\Rightarrow (z_1 - z_2)'' + p(x) (z_1 - z_2)' + q(x) (z_1 - z_2) = f(x) - f(x) = 0$$

$$\Rightarrow$$

Theorem

Let $y = y_1(x)$ and $y = y_2(x)$ be linearly independent solutions of the reduced equation (5) and let $z = z(x)$ be a particular solution of (4). If $u = u(x)$ is any solution of (4), then there exist constants C_1 and C_2 such that

$$u(x) = C_1 y_1(x) + C_2 y_2(x) + z(x).$$

$$y'' + P(x)y' + Q(x)y = f \quad \text{—— (N)}$$

$$y'' + P(x)y' + Q(x)y = 0 \quad \text{—— (H)}$$

y_1 y_2

$u \longrightarrow (N)$

$z \longrightarrow (N)$

$u - z \longrightarrow (H)$

Sol ves.

$u - z = C_1 y_1 + C_2 y_2$

$\Rightarrow u = z + (C_1 y_1 + C_2 y_2)$

General Solution for Nonhomogeneous Linear ODE

• Let $y = y_1(x)$ and $y = y_2(x)$ be linearly independent solutions of the reduced equation (5) and let $z = z(x)$ be a particular solution of (4). If $u = u(x)$ is any solution of (4), then there exist constants C_1 and C_2 such that

$$u(x) = C_1 y_1(x) + C_2 y_2(x) + z(x).$$

• Let $y = y_1(x)$ and $y = y_2(x)$ be linearly independent solutions of the reduced equation (5) and let $z = z(x)$ be a particular solution of (4). If $u = u(x)$ is any solution of (4), then there exist constants C_1 and C_2 such that

$$\underbrace{u(x)}_{\text{general solution of (4)}} = \underbrace{C_1 y_1(x) + C_2 y_2(x)}_{\text{general solution of (5)}} + \underbrace{z(x)}_{\text{particular solution of (4)}}.$$

Superposition Principle

The following result is sometimes useful in finding particular solutions of nonhomogeneous equations. It is known as the superposition principle.

Theorem

Given the second-order linear nonhomogeneous equation

$$y'' + p(x)y' + q(x)y = f(x) + g(x) = h(x) \tag{6}$$

If $z = z_f(x)$ and $z = z_g(x)$ are particular solutions of

$$y'' + p(x)y' + q(x)y = f(x) \quad \text{and} \quad y'' + p(x)y' + q(x)y = g(x),$$

respectively, then $z(x) = z_f(x) + z_g(x)$ is a particular solution of (6).

$z_f'' + p(x) z_f' + q(x) z_f = f(x)$ and $z_g'' + p(x) z_g' + q(x) z_g = g(x)$

$+ \Rightarrow (z_f + z_g)'' + p(x)(z_f + z_g)' + q(x)(z_f + z_g) = f(x) + g(x)$

Variation of Parameters:- To find the general

sol^n of

$$y'' + p(x)\,y' + q(x)\,y = f(x) \quad \text{——} \quad (N)$$

we need to find:

(i) a linearly independent pair of sol^{n}s y_1, y_2 of the reduced equation

$$y'' + p(x)\,y' + q(x)\,y = 0. \quad \text{——} \quad (H).$$

(ii) A particular sol^n z of (N).

- The method of variation of parameters uses a pair of linearly independent solⁿs of the reduced equation to construct a particular solⁿ of (N).

- Let $y_1(x)$ and $y_2(x)$ be linearly independent solⁿ of the reduced equation (H). Then,

$$y = c_1 y_1(x) + c_2 y_2(x)$$

is the general solⁿ of (H). We replace the arbitrary constants by functions $\boxed{u = u(x)}$ and $\boxed{v = v(x)}$, which are to be determined so that

$$z(x) = u(x)\, y_1(x) + v(x)\, y_2(x)$$

is a particular solⁿ of the non homogeneous equation (N).

$$Z(x) = u(x)\, y_1(x) + v(x)\, y_2(x)$$

$$\Rightarrow z'(x) = u'(x)\, y_1(x) + u(x)\, y_1'(x) + v'(x)\, y_2(x) + v(x)\, y_2'(x)$$

$$= u(x)\, y_1'(x) + v(x)\, y_2'(x)$$ by choosing $\boxed{u'(x)\, y_1(x) + v'(x)\, y_2(x) = 0}$

$$z''(x) = u'(x)\, y_1'(x) + u(x)\, y_1''(x) + v'(x)\, y_2'(x) + v(x)\, y_2''(x)$$

z will satisfy

$$u'(x)\, y_1'(x) + u(x)\, y_1''(x) + v'(x)\, y_2'(x) + v(x)\, y_2''(x) + p(x)\big(u(x)\, y_1'(x) + v(x)\, y_2'(x)\big) + q(x)\big(u(x)\, y_1(x) + v(x)\, y_2(x)\big) = f(x)$$

$$\Rightarrow u(x)\underbrace{\big(y_1''(x) + p(x)\, y_1'(x) + q(x)\, y_1(x)\big)}_{=0} + v(x)\underbrace{\big(y_2''(x) + p(x)\, y_2' + q(x)\, y_2(x)\big)}_{0} + u'(x)\, y_1'(x) + v'(x)\, y_2'(x) = f(x)$$

$$\Rightarrow \boxed{u'(x)\, y_1'(x) + v'(x)\, y_2'(x) = f(x)}$$

$$\boxed{\begin{aligned} u'(x)\, y_1(x) + v'(x)\, y_2(x) &= 0 \\ u'(x)\, y_1'(x) + v'(x)\, y_2'(x) &= f(x) \end{aligned}}$$

- The replacement of the parameters c_1 and c_2 by the "variables" u and v is the basis for the term "variation of parameters".
- Since there are two unknowns u and v to be determined, we shall impose two conditions on these unknowns:
  - (a) z should solve the differential equation
  - (b) we shall choose it in a manner that will simplify our calculations.

- Solving for u' and v', we find

$$u' = \frac{-y_2 f}{y_1 y_2' - y_1' y_2} \quad \text{and} \quad v' = \frac{y_1 f}{y_1 y_2' - y_1' y_2}$$

- Since $y_1 y_2' - y_1' y_2 = W$ and is nonzero as y_1 and y_2 are linearly independent solⁿs. of the reduced equation (H).

- Finally, $u = -\int \frac{y_2 f}{y_1 y_2' - y_1' y_2} dx$ and $v = \int \frac{y_1 f \, dx}{y_1 y_2' - y_1' y_2}$

$$= \int \frac{y_2(x) f(x)}{W[y_1, y_2](x)} dx \qquad = \int \frac{y_1(x) f(x)}{W[y_1, y_2](x)} dx$$

Use of a known solⁿ to find another

If we know one solⁿ of
$$y'' + p(x)y' + q(x)y = 0$$
we can find another solⁿ y_2 which is independent of y_1 (i.e., $y_1'' + p(x)y_1' + q(x)y_1 = 0$).

Suppose y_1 is a non-zero solⁿ. Assume $y_2 = vy_1$ is another solⁿ. (y_2 must satisfy : $\underline{y_2'' + p(x)y_2' + q(x)y_2 = 0}$)

On substituting
$$y_2 = vy_1, \quad y_2' = vy_1' + v'y_1 \quad \& \quad y_2'' = vy_1'' + 2v'y_1' + v''y_1$$

into the (H) and rearranging, we get

$$v\overbrace{\left(y_1'' + py_1' + qy_1\right)}^{0} + v''y_1 + v'(2y_1' + py_1) = 0$$

Since y_1 is a sol^n, we deduce

$$v''y_1 + v'(2y_1' + py_1) = 0$$

$$\Rightarrow \frac{v''}{v'} = -\frac{2y_1'}{y_1} - p$$

By integrating, we get

$$-\log v' = -2\log y_1 - \int p\,dx$$

$$\Rightarrow v' = \frac{1}{y_1^2} e^{-\int p(x)\,dx}$$

$$\Rightarrow v = \int \frac{1}{y_1^2} e^{-\int p(x)\,dx}$$

$$\boxed{y_2 = y_1\left(\int \frac{1}{y_1^2} e^{-\int p(x)\,dx}\right)}$$

$$y'' + a_1 y' + b_1 y = 0. \qquad (r^2 + a_1 r + b_1) = 0$$

$\boxed{P_1(x) = a_1}$ — $r_1 \neq r_2$ (r_1, r_2 real)

In case of repeated roots:

$y_1(x) = e^{ax}$ is one sol^n.

Qn. How to find another sol^n from this y_1?

$$\underbrace{r^2 - 2ar + a^2 = 0}_{\text{ODE}}$$

$$y'' - 2ay' + a^2 y = 0$$

$$y'' + P(x)y' + q(x)y = 0$$

$$P(x) = -2a$$

$$q(x) = a^2$$

$$v(x) = \int \frac{1}{e^{2ax}} \left(e^{+\int \boxed{2a} dx} \right) dx$$

$$= \int \frac{1}{\cancel{e^{2ax}}} \cancel{e^{2ax}} \, dx$$

$$= \boxed{x}$$

$$y_2(x) = y_1(x) v(x) = \underline{x e^{ax}}$$

Gen Sol^n: $y(x) = c_1 e^{ax} + c_2 x e^{ax}$

Thank You

Questions?